SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:100**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I S.Shouib Akthar (192312208) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

S. Shuib Akthar
Signature of the student:

Q. No	*Understanding of Concepts (25)*	*Experimental Design (30)*	*Use of Tools (20)*	*Analysis of Results (15)*	*Documentation (10)*	*Total Score (out of 100)*
1	5 /5	5 /6	4 /4	2 /3	1 /2	17
2	5 /5	5 /6	4 /4	2 /3	1 /2	17
3	5 /5	5 /6	4 /4	2 /3	1 /2	17
4	5 /5	5 /6	4 /4	2 /3	1 /2	17
5	5 /5	5 /6	4 /4	2 /3	1 /2	17
OVERALL RUBRICS VALUE						
	5 /5	5 /5	4 /4	2 /4	1 /2	
	25/25	25/30	20/20	10/15	5/10	85 /100

Annexure A

APPLICATION- BASED QUESTIONS

Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
Deduce the most likely original message.
Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
Demonstrate how the encryption process transforms the message step-by-step.
Explain how the repeating keyword influences each character of the ciphertext.

Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
Analyze how investigators could detect the presence of hidden data within the image.
Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CO1 ASSESSMENT

TOOL - 1

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Course Code: CSA5117

Course Name: Cryptography and Network Security

ENS5H7
CSA5117

CRYPTOGRAGHY AND NETWORK SECURITY

CO1- AT1

① * Event Communication Security

A Student organization is preparing to Share Confidential event details with its core members using a simple classical encryption method.

a) Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

b) Explain how an authorized member, aware of the encoding rule, would retrieve the original message

c) Briefly explain the working of Caesar Cipher.

a) Encryption:

Plain Text : MEET AT NOON , Shift = +4

plain Text	M	E	E	T	A	T	N	O	O	N
Cipher Text	Q	I	I	X	E	X	R	S	S	R

Encrypted Message: QIIX EX RSSR

b) Decryption:

The receiver subtracts 4 from each encrypted letter.

Q → M
I → E
X → T
R → N
S → O

Original Message : MEET AT NOON

c) Explaination:

The Caesar Cipher is a substitution cipher where each letter is shifted by a fixed no.of positions. Here, every letter is shifted by 4 positions. The receiver decrypts the message by shifting each letter 4 positions backward. Although simple and easy to implement, it is vulnerable to brute-force attacks because there are only 25 possible keys.

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(2) * Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring.

a) Deduce the most likely original message

b) Justify your answer by analyzing repeating patterns, letter structures, or known cipher characterstics.

a) Original Message

The intercepted message is encrypted using the Atbash Cipher.

THIS IS A SECRET

b) Justification:

The Atbash Cipher replaces every alphabet with is opposite alphabet:

A ⟷ Z

B ⟷ Y

C ⟷ X

D ⟷ W

The repeated word RH becomes IS, which is a common English word. The final word HVXIVG becomes SECRET, making the sentence meaningful. The letter patterns and word lengths match normal English, confirming that the message is "THIS IS A SECRET".

③ * Secure Messaging System

A Company deploys a keyword-based encryption technique for internal Communication. An employee sends the word "HELLO" using the keyword "KEY."

a) Demonstrate how the encryption process transforms the message step-by-step.

b) Explain the repeating keyword influences each character of the Ciphertext.

a) Encryption Process:

Plaintext : HELLO

Keyword : KEY , Repeated keyword: KEYKE

plain text	H	E	L	L	O
keyword	K	E	Y	K	E
Cipher text	R	I	J	V	S

Encrypted Message : RIJVS

b) Explaination:

The keyword repeats until it matches the length of the plaintext. Each keyword letter provides a different Shift value, making every plaintext character encrypt differently. This increases security compared to the Caesar Cipher because the shift is not constant.

④ Suspicious Digital Asset Investigation

* During a digital forensic audit, investigators suspect that a Company logo file contains hidden Confidential information.

a) Analyze how investigators could detect hidden data within the image.

b) Recommend two techniques to strengthen the security of such hidden communications against detection or extraction.

a) Detecting Hidden Data:

* Investigators can detect hidden data by:

* Performing Least Significant Bit (LSB) analysis. Using Steganalysis Software such as StegExpose or StegSolve

* Examining image metadata

* Comparing the suspected image with the original image.

b) Security Improvements

Technique 1: Encrypt the Secret message using AES Encryption before embedding it into the image.

Technique 2: Use random pixel embedding Controlled by a Secret key instead of embedding data Sequentially.

these methods make hidden information much harder to detect & extract.

(5) Legacy Encryption in Corporate Communication:

A legacy System uses a transposition based encryption method with 3 levels of arrangement. The message is:

"WECRLTEERDSOEEFEAOCAIVDEN"

a) Determine the most Probable original message by reversing the encryption process.

b) Critically evaluate why Such a method is Vulnerable in modern Communication Systems.

c) Propose one practical enhancement to improve the Confidentiality of messages.

a) Original Message:

Using the 3-Rail Fence cipher, the decrypted message is

WE ARE DISCOVERED FLEE AT ONCE

b) Vulnerability:

The Rail Fence Cipher is vulnerable because it only rearranges the letters without changing them. Letter frequencies remain the same, making it easy to crack using brute-force methods or pattern analysis. Modern computers can decrypt such ciphers quickly.

c) Practical Enhancement:

A practical improvement is to use AES (Advanced Encryption Standard) instead of a simple transposition cipher. AES uses a secret key and multiple rounds of encryption, providing strong protection against modern crypto-graphic attacks.